Why Fault Detectors Learn the Wrong Rule: Conjunctive Signatures, Composite Nulls, and the Supervision They Need

Theory notes with complete proofs and numerical checks (draft)

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Abstract

A detector trained on normal versus one fault reaches near-perfect detection, yet in a fleet it flags many units that have a different fault or none. We model a multi-sensor fault as a *conjunction*: k signature channels shift together, at the same time. We prove five things. (i) In the overlapping-class regime the detector that is optimal for the training task is *additive* across channels, does not depend on class prevalence, and is what every cross-entropy-consistent learner converges to; more data converges *to* the shortcut. (ii) Its false-alarm rate on any partially overlapping fault follows a closed-form law that depends only on the target's detectability and the share of its signal the other fault reproduces. A corollary gives a *detectability paradox*: the more reliably the target is detected, the more surely every partial look-alike is flagged. The resulting trade-off curve predicts the partial-match rates of the toy study's MLP and LSTM from their detection rate alone (0.72 vs. 0.73; 0.19 vs. 0.19–0.20). (iii) Deployment is a test against a *composite* null (normal or any fault that moves a strict subset of the signature). There, additive detectors lose a $\sqrt{k}$ factor in effective signal, or all power when overlap is unbounded; an intersection–union ("capped evidence") detector is valid and within a factor k of minimax. (iv) In the separated regime the choice is made by implicit bias: any norm-bounded network places near-misses at least $1 - \kappa/\sqrt{k}$ of the way from normal to fault, and rejecting all k near-misses costs at least k times the minimum path norm. This yields a necessary fraction of near-miss examples that grows with weight decay, roughly linearly for small weight decay. (v) The minimal supervision that identifies the rule is k channel near-misses plus $\binom{k}{2}$ temporal near-misses. The temporal count follows from Helly's theorem on the line, and it implies that fully staggered negatives teach nothing about simultaneity. All claims are checked numerically (Section 9).

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1 Results at a glance

Result	Statement (informal)	Checked by
Thm 3.1	Training-optimal detector is additive; prevalence- and n-invariant; consistent learners converge to it	C1 (law holds for $n \geq 2000$, all prevalences)
Thm 3.3, Cor 3.4	Partial-overlap false-alarm law; power–specificity curve; detectability paradox	C1, C2, C8 (toy slide 12)
Thm 4.1, Prop 4.2	Composite null: additive loses $\sqrt{k}$ or everything; intersection–union near-minimax; capped evidence is necessary	C3
Thm 5.2, 5.4	Near-miss position $\geq 1 - \kappa\rho$; rejecting near-misses costs $\geq k\cdot$ path norm	C4
Prop 5.6	Necessary near-miss fraction $p^*(\lambda)$, increasing in λ	C9, C5
Lemma 6.1, 6.2	Window statistics are blind to timing and to sign	C6a
Lemma 6.3	Temporal dependence enters only through σ_{eff}	C7
Thm 7.1, 7.2	Minimal teaching set: k channel + $\binom{k}{2}$ temporal near-misses	C6b
Prop 7.5	How many examples per near-miss type	—

Corrections to earlier slides. (1) "Adding per-sensor evidence cannot express the AND" is true only for *uncapped* evidence; a generalized additive model with bounded per-channel terms can represent it (Prop. 4.2). (2) The $\delta/(2\sqrt{k})$ margin statement applies to *linear* additive detectors.

2 Setting

A window is $X \in \mathbb{R}^{T\times C}$ (T time steps, C channels). Normal windows are $X = W$ with W a stationary noise process. The target fault has signature $S \subseteq [C]$, $|S| = k \geq 2$, directions $s_j \in \{\pm 1\}$, magnitudes $\delta_j > 0$ and an active interval $I \subseteq [T]$:

$$X = W + \textstyle\sum_{j\in S} s_j\delta_j\, \mathbf{1}_I\, e_j^\top.$$

Shift statistic. For a baseline set $B \subset [T]$ disjoint from I, let $a_j(X) = s_j\big(\frac{1}{|I|}\sum_{t\in I} X_{tj} - \frac{1}{|B|}\sum_{t\in B} X_{tj}\big)$ (with $s_j = +1$ for $j \notin S$).

Assumption 2.1 (Gaussian shift)**.** W is a stationary Gaussian process. Then $a \sim N(0, \Sigma)$ under normal and $a \sim N(\boldsymbol{\delta}, \Sigma)$ under the target, with $\boldsymbol{\delta} = (\delta_j \mathbf{1}[j \in S])_j$ and Σ determined by the autocovariance of W (Lemma 6.3).

Sections 3–4 work with a (the onset is treated as known); Sections 5–7 also cover raw windows.

Other faults. A fault with shift vector μ produces $a \sim N(\mu, \Sigma)$. Special cases: the m-of-k *partial* fault $\mu = \boldsymbol{\delta} \odot \mathbf{1}_{S'}$ with $S' \subset S$, $|S'| = m$; the *near-miss* $m = k - 1$; *BIG* $\mu = c\,\delta_j e_j$; *flip* $\mu = -\boldsymbol{\delta}$; *off-signature* μ supported on S^c.

Notation. Φ is the standard normal CDF, $z_\alpha = \Phi^{-1}(1-\alpha)$, $D^2 = \boldsymbol{\delta}^\top \Sigma^{-1} \boldsymbol{\delta}$ (target detectability), and

$$r(\mu) = \frac{\boldsymbol{\delta}^\top \Sigma^{-1} \mu}{D^2} \qquad \text{(share of the target's signal that } \mu \text{ reproduces).}$$

A detector is α-*calibrated* if it flags a fraction α of normal windows (as in all experiments).

3 Regime I (overlapping classes): the shortcut is optimal for the training task

Theorem 3.1 (Training-optimal detector)**.** *Under Assumption 2.1, let the training prevalence of the target be $\pi \in (0,1)$.*

(i) The Bayes log-odds is $\ell_\pi(a) = \boldsymbol{\delta}^\top \Sigma^{-1} a - \frac{1}{2} D^2 + \log \frac{\pi}{1-\pi}$, which is additive (linear) in a.

(ii) For every $\alpha \in (0,1)$ the most powerful size-α test of normal against target is $\phi^*(a) = \mathbf{1}\{\boldsymbol{\delta}^\top \Sigma^{-1} a > z_\alpha D\}$. It does not depend on π.

(iii) Let $\hat{f}_n$ be any learner with $\hat{f}_n \to \ell_\pi$ in probability under the training mixture $P_\pi = (1-\pi)N(0,\Sigma) + \pi N(\boldsymbol{\delta}, \Sigma)$. This includes gradient-descent logistic regression, whose model is well specified by (i), and any cross-entropy-consistent flexible learner. Let $\hat{\phi}_n = \mathbf{1}\{\hat{f}_n > \hat{t}_n\}$ with $\hat{t}_n$ the $(1-\alpha)$-quantile of $\hat{f}_n$ under $N(0,\Sigma)$. Then for every distribution Q absolutely continuous with respect to Lebesgue measure, $Q(\hat{\phi}_n \neq \phi^*) \to 0$.

Proof. (i) The ratio of the two Gaussian densities with common covariance is $\exp(\boldsymbol{\delta}^\top \Sigma^{-1} a - \frac{1}{2} \boldsymbol{\delta}^\top \Sigma^{-1} \boldsymbol{\delta})$. Bayes' rule adds $\log \frac{\pi}{1-\pi}$.

(ii) By the Neyman–Pearson lemma, the most powerful test rejects for large likelihood ratio, which is increasing in $u = \boldsymbol{\delta}^\top \Sigma^{-1} a$. Under the null $u \sim N(0, D^2)$, so the size-α threshold is $z_\alpha D$. The prior does not appear.

(iii) Write t^* for the $(1-\alpha)$-quantile of ℓ_π under $P_0 = N(0,\Sigma)$; then $\{\ell_\pi > t^*\} = \{\phi^* = 1\}$ by (ii). Since $P_0 \leq P_\pi/(1-\pi)$, convergence in P_π-probability implies convergence in P_0-probability. Under P_0, ℓ_π is Gaussian with a continuous, strictly increasing CDF, so the quantiles converge: $\hat{t}_n \to t^*$. Fix $\varepsilon > 0$. The disagreement set is contained in

$$\{|\ell_\pi - t^*| \leq 2\varepsilon\} \;\cup\; \{|\hat{f}_n - \ell_\pi| > \varepsilon\} \;\cup\; \{|\hat{t}_n - t^*| > \varepsilon\}.$$

The third event is deterministic and eventually empty. Q is absolutely continuous and P_π has an everywhere-positive density, so $Q \ll P_\pi$. For a finite measure this gives, for every $\eta > 0$, a $\gamma > 0$ with $P_\pi(A) < \gamma \Rightarrow Q(A) < \eta$; hence $Q(|\hat{f}_n - \ell_\pi| > \varepsilon) \to 0$. Finally $Q(|\ell_\pi - t^*| \leq 2\varepsilon) \to 0$ as $\varepsilon \to 0$, because ℓ_π is a non-degenerate affine function of a and so has no atom under Q. □

Remark 3.2 (Interpretation). In this regime the "broad" rule is not a learning failure. It is the unique optimum of the training problem, and more data, rebalancing or longer training make a learner converge to it. Theorem 3.1 extends to exponential-family noise with channel-separable sufficient statistics: the log-odds is still additive in the sufficient statistics.

Theorem 3.3 (Partial-overlap false-alarm law)**.** *Under Assumption 2.1, the training-optimal detector ϕ^* flags a fault with shift μ with probability*

$$\Pr_\mu(\phi^* = 1) = \Phi\big(D\, r(\mu) - z_\alpha\big).$$

In particular, with independent signature channels of equal noise σ and equal shifts δ ($D = \delta\sqrt{k}/\sigma$):

$$\textit{m-of-k partial: } \Phi\Big(\tfrac{m\delta}{\sigma\sqrt{k}} - z_\alpha\Big), \qquad \textit{BIG ($c\delta$ on one channel): } \Phi\Big(\tfrac{c\delta}{\sigma\sqrt{k}} - z_\alpha\Big),$$
$$\textit{flip: } \Phi(-D - z_\alpha), \qquad \textit{off-signature: } \alpha.$$

Proof. Under $N(\mu, \Sigma)$ the statistic $u = \boldsymbol{\delta}^\top \Sigma^{-1} a$ is Gaussian with mean $\boldsymbol{\delta}^\top \Sigma^{-1}\mu = D^2 r(\mu)$ and variance $\boldsymbol{\delta}^\top \Sigma^{-1}\Sigma\Sigma^{-1}\boldsymbol{\delta} = D^2$. Hence $\Pr(u > z_\alpha D) = \Phi(Dr(\mu) - z_\alpha)$. The special cases follow from $r = m/k$, c/k, -1 and 0. □

Corollary 3.4 (Power–specificity curve and the detectability paradox)**.** *Let $\beta = \Phi(D - z_\alpha)$ be the detection rate of ϕ^*. Then for every μ*

$$\Pr_\mu(\phi^* = 1) = \Phi\Big(r(\mu)\big(\Phi^{-1}(\beta) + z_\alpha\big) - z_\alpha\Big).$$

For every partial fault with $r(\mu) \in (0, 1)$ this is strictly increasing in β and tends to 1 as $\beta \to 1$. Any change that makes the target easier to detect at a fixed false-alarm budget therefore increases false alarms on every *partially overlapping fault: larger shifts, less noise, longer windows, or weaker temporal correlation (Lemma 6.3).*

Proof. Substitute $D = \Phi^{-1}(\beta) + z_\alpha$ into Theorem 3.3. Monotonicity holds because $r > 0$ and Φ is increasing. □

Corollary 3.5 (Loudest-channel dominance)**.** *With independent channels, a fault that reproduces only channel j is flagged with probability $\Phi(Dr_j - z_\alpha)$, where $r_j = (\delta_j/\sigma_j)^2/D^2$ is channel j's share of the target SNR. Faults that hit the loudest signature channel are flagged most; the near-miss that drops the loudest channel is flagged least.*

Remark 3.6 (Model-agnostic prediction). If a trained model's score is close to an increasing function of the additive statistic, Corollary 3.4 predicts its partial-match rate from its own detection rate, with no fitting. On the toy study (check C8) this gives P2 0.72 vs. 0.73 and P1 0.19 vs. 0.19–0.20 for the MLP and LSTM. Window-statistics models exceed the curve (P2 0.83–0.89 vs. 0.78), consistent with their additional timing blindness (Lemma 6.1).

Remark 3.7 (Unknown onset). Without an oracle onset, the likelihood-ratio detector scans candidate onsets, $\max_\tau \boldsymbol{\delta}^\top \Sigma^{-1} a(\tau)$. It is still additive across channels at every τ, so partial faults with $r > 0$ raise every scanned statistic. The exact law becomes a scan-statistic tail and is left open (Section 12).

4 What deployment needs: a composite null

In a fleet, "not the target" means normal *or* any other fault. The faults that matter move a strict subset of the signature. Take independent signature channels with noise σ_j, detectors that depend on a_S only, and fault-direction shifts. Define the *overlap null*

$$\mathcal{N}_{k-1}(M) = \big\{N(\mu, \Sigma) : \ \mathrm{supp}(\mu) \subseteq S' \subsetneq S, \ 0 \le \mu_j \le M_j\big\}, \qquad M \in (0, \infty]^k.$$

A detector is *deployment-valid* at level α if $\sup_{Q \in \{P_0\} \cup \mathcal{N}_{k-1}(M)} Q(\phi = 1) \le \alpha$.

Theorem 4.1 (Composite null)**.** *Let $M_j \ge \delta_j$ for all j.*

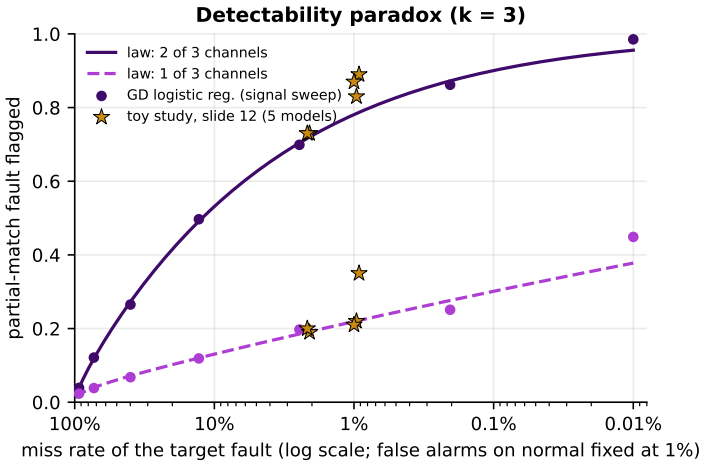


Figure 1: Power–specificity curve (Corollary 3.4), $k = 3$, 1% false alarms on normal. Lines: the law, no fitted parameters. Circles: gradient-descent logistic regression over a signal-strength sweep (check C2). Stars: the five models of the toy study at their own detection rates (check C8).

(a) (Upper bound.) *Every deployment-valid level-α test has target power at most* $\min_{j\in S}\Phi(\delta_j/\sigma_j - z_\alpha)$.

(b) (Intersection–union attains it up to a factor k.) $\phi_{IU}(a) = \mathbf{1}\{\min_{j\in S} a_j/\sigma_j > z_\alpha\}$ *is deployment-valid for* $M = \infty$. *Its power is* $\prod_j \Phi(\delta_j/\sigma_j - z_\alpha)$, *so its miss probability is at most* $\sum_j \Phi(z_\alpha - \delta_j/\sigma_j) \le k \cdot \max_j \Phi(z_\alpha - \delta_j/\sigma_j)$, *i.e. at most k times the minimax lower bound on the miss probability from (a).*

(c) (Additive, bounded overlap.) *With equal δ, σ and $M = \delta$, every deployment-valid linear detector* $\mathbf{1}\{w^\top a_S > t\}$ *with $w \ge 0$ has power at most* $\Phi(\delta/(\sigma\sqrt{k}) - z_\alpha)$, *with equality at* $w = \mathbf{1}$.

(d) (Additive, unbounded overlap.) *If $M = \infty$, every deployment-valid linear detector with $w \ge 0$, $w \ne 0$ has power* 0.

(e) (What training produces.) *The training-optimal detector ϕ^* is not deployment-valid: its false-alarm rate on the worst near-miss is* $\Phi\big((k-1)\delta/(\sigma\sqrt{k}) - z_\alpha\big) > \alpha$ *(equal case).*

Proof. (a) Fix j and let $\mu^{(j)} = \boldsymbol{\delta} - \delta_j e_j \in \mathcal{N}_{k-1}(M)$. A deployment-valid ϕ is in particular a level-α test of $N(\mu^{(j)}, \Sigma)$ against $N(\boldsymbol{\delta}, \Sigma)$. These two laws differ only in the mean of coordinate j, by δ_j, so by Neyman–Pearson the most powerful level-α test has power $\Phi(\delta_j/\sigma_j - z_\alpha)$. Take the minimum over j.

(b) Size (Berger, 1982): for $Q \in \mathcal{N}_{k-1}(\infty)$ some $j \in S \setminus S'$ has $\mu_j = 0$, so $Q(\phi_{IU} = 1) \le Q(a_j > z_\alpha\sigma_j) = \alpha$. Under P_0 the same bound holds. Power: the channels are independent and each event $\{a_j > z_\alpha\sigma_j\}$ has probability $\Phi(\delta_j/\sigma_j - z_\alpha)$ under the target. The miss bound follows from $1 - \prod_j(1 - x_j) \le \sum_j x_j$.

(c) Write $\|w\| = \|w\|_2$. Validity on $\mu = \delta(\mathbf{1} - e_i)$ for each i requires $t \ge \delta\sum_{j\ne i} w_j + z_\alpha\sigma\|w\|$. The binding constraint is $i = \arg\min_j w_j$, so $t \ge \delta(\|w\|_1 - \min_j w_j) + z_\alpha\sigma\|w\|$. The power is $\Phi\big((\delta\|w\|_1 - t)/(\sigma\|w\|)\big) \le \Phi\big(\delta\min_j w_j/(\sigma\|w\|) - z_\alpha\big)$. Finally $\min_j w_j \le \|w\|/\sqrt{k}$, with equality iff $w \propto \mathbf{1}$.

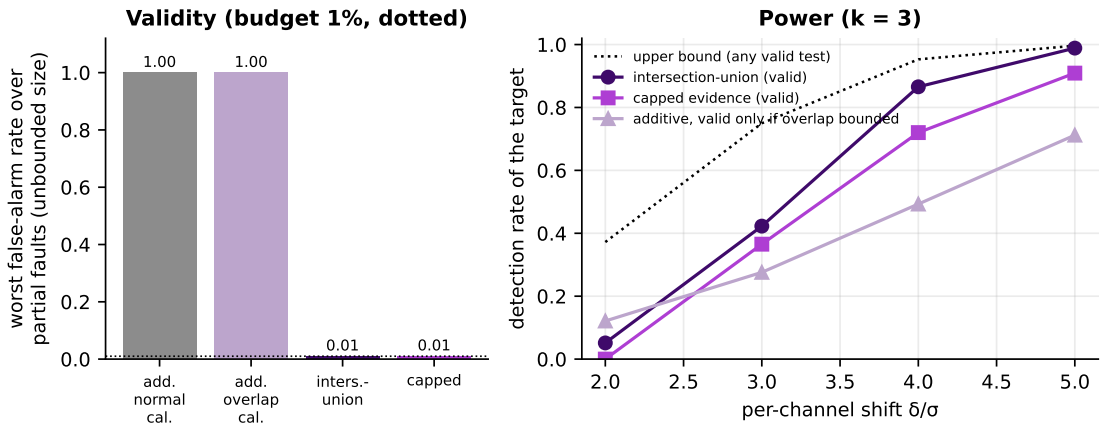


Figure 2: Composite null, $k = 3$ (check C3). Left: worst false-alarm rate over partial faults of unbounded size. Detectors that add uncapped evidence are invalid. Right: target detection rate of the valid detectors and of the additive detector that is valid only when overlaps are bounded by the target's own shift.

(d) Pick j with $w_j > 0$. Since $k \geq 2$, some $S' \ni j$ with $|S'| = k-1$ exists, so $\mu = Ae_j \in \mathcal{N}_{k-1}(\infty)$ for every $A > 0$. Under $N(\mu, \Sigma)$, $w^\top a_S \sim N(w_j A, \sigma^2 \|w\|^2)$, so the false-alarm rate tends to 1 as $A \to \infty$ for any finite t. Validity forces $t = +\infty$, hence zero power.

(e) Theorem 3.3 with $r = (k-1)/k$. □

Proposition 4.2 (Capped evidence is necessary and sufficient)**.** *Consider generalized additive detectors* $\mathbf{1}\{\sum_{j \in S} g_j(a_j) > t\}$ *with nondecreasing* g_j*, under* $M = \infty$*.*

(a) If some g_j *is unbounded above, every deployment-valid such detector has power* 0*.*

(b) If every g_j *is bounded, the class contains* ϕ_{IU} *(take* $g_j = \mathbf{1}\{a_j > z_\alpha \sigma_j\}$ *and* $t = k - \frac{1}{2}$*), so it attains Theorem 4.1(b).*

The same holds for channel-independent sequence models $f(X) = \sum_j h_j(X_{\cdot j})$*, the structure of per-channel encoders with a linear head: such a model is deployment-valid with nontrivial power only if each* h_j *is bounded along fault-direction shifts.*

Proof. (a) Repeat the proof of Theorem 4.1(d) with $g_j(a_j) \to \infty$ in place of $w_j a_j \to \infty$. The other summands are fixed in distribution, so the false-alarm rate tends to 1 for any finite t. (b) is the construction stated. For the sequence-model version, apply (a) to h_j evaluated along $X + A\, s_j \mathbf{1}_I e_j^\top$, $A \to \infty$. □

Remark 4.3. Proposition 4.2 replaces the earlier slide statement. The channel-independent model in our experiments uses ReLU encoders with a linear head and is unbounded; it flags one-loud-channel faults at 0.99–1.00 and stays at 0.67 on near-misses after the fix. The theory predicts that a *capped* version (bounded per-channel output) can be fixed. This is now an experiment in the pipeline.

5 Regime II (separated classes): what gradient descent picks

When normal and fault are well separated (our real-data targets: one sensor alone gives 89–99.8% training accuracy), near-misses have negligible training density and Theorem 3.1(iii) no longer

pins down the detector there. Many rules fit the training data. Gradient descent on homogeneous networks with logistic loss converges in direction to a KKT point of the minimum-norm (maximum-margin) problem (Lyu and Li, 2020; Ji and Telgarsky, 2020), and weight decay explicitly penalises the norm. Both favour small norm, and small norm bounds the Lipschitz constant.

Lemma 5.1 (Norm bounds Lipschitz)**.** *For $f(x) = \sum_i v_i\,\sigma(w_i^\top x + b_i) + c$ with σ 1-Lipschitz, $|f(x) - f(y)| \le R(\theta)\|x - y\|$ with path norm $R(\theta) = \sum_i |v_i|\|w_i\|$. Moreover $\min_{\text{rescalings}} \frac{1}{2}\sum_i (v_i^2 + \|w_i\|^2 + b_i^2) \ge R(\theta)$. For depth-$L$ networks, $\mathrm{Lip}(f) \le \prod_l \|W_l\|_{op} \le (\|\theta\|^2/L)^{L/2}$.*

Proof. $|f(x) - f(y)| \le \sum_i |v_i||w_i^\top(x-y)| \le \sum_i |v_i|\|w_i\|\|x-y\|$. For the second claim, $(v, w, b) \mapsto (v/\gamma, \gamma w, \gamma b)$ leaves f unchanged, and $\min_\gamma \frac{1}{2}(v^2/\gamma^2 + \gamma^2(\|w\|^2 + b^2)) = |v|\sqrt{\|w\|^2 + b^2} \ge |v|\|w\|$. The deep case is submultiplicativity followed by AM–GM. □

Theorem 5.2 (Near-miss position)**.** *Let x_0, x_+ be normal and fault prototypes with $\Delta = \|x_+ - x_0\|$, and let f satisfy $f(x_+) - f(x_0) = G > 0$. Let $R_{\min} = G/\Delta$, the smallest Lipschitz constant compatible with this separation, which a single unit attains, and let $\kappa = \mathrm{Lip}(f)/R_{\min} \ge 1$. For any x with $\|x - x_+\| = \rho\Delta$,*

$$p(x) := \frac{f(x) - f(x_0)}{f(x_+) - f(x_0)} \;\ge\; 1 - \kappa\rho.$$

Proof. $f(x) \ge f(x_+) - \mathrm{Lip}(f)\rho\Delta = f(x_+) - \kappa\rho G$. Subtract $f(x_0)$ and divide by G. □

Corollary 5.3 (Channel and temporal near-misses)**.** *(a) In shift-statistic space with equal shifts, a channel near-miss has $\rho = 1/\sqrt{k}$, so $p \ge 1 - \kappa/\sqrt{k}$. In raw windows where each signature channel is shifted by δ on L steps, ρ is again $\sqrt{L}\delta/(\sqrt{kL}\delta) = 1/\sqrt{k}$.*

(b) A temporal near-miss that displaces one channel's shift block by $\Delta t \le L$ steps has $\|x - x_+\| = \delta\sqrt{2\Delta t}$, so $\rho = \sqrt{2\Delta t/(kL)}$ and $p \ge 1 - \kappa\sqrt{2\Delta t/(kL)}$. A low-norm detector cannot separate the target from a staggering of fewer than about $kL/(8\kappa^2)$ steps.

An α-calibrated threshold sits near the normal prototype, at some p_α small when noise is small. Any near-miss with $p(x) > p_\alpha$ is flagged.

Theorem 5.4 (The narrow rule costs k times the norm)**.** *Let f be a two-layer network as in Lemma 5.1 with ReLU activation, and let $x^{(j)} = x_+ - \delta\,\mathbf{1}_{\mathcal{B}} e_j^\top$ be the channel-j near-miss, where $\mathcal{B}$ is the shift block of length L ($L = 1$ in shift-statistic space). If $f(x_+) - f(x^{(j)}) \ge g > 0$ for every $j \in S$, then*

$$R(\theta) \;\ge\; \frac{\sqrt{k}\,g}{\delta\sqrt{L}}, \qquad \text{so} \qquad \frac{R(\theta)}{R_{\min}} \;\ge\; k\,\frac{g}{G}.$$

For unequal shifts, $R(\theta) \ge g\sum_{j\in S}\delta_j^{-1}/\sqrt{kL}$.

Proof. ReLU is 1-Lipschitz, so for each unit $|\mathrm{relu}(w_i^\top x_+ + b_i) - \mathrm{relu}(w_i^\top x^{(j)} + b_i)| \le \delta\sum_{t\in\mathcal{B}}|w_{i,tj}|$. Hence $g \le \delta\sum_i |v_i|\sum_{t\in\mathcal{B}}|w_{i,tj}|$ for each j. Summing over $j \in S$ and applying Cauchy–Schwarz to the kL coordinates in $\mathcal{B}\times S$,

$$kg \;\le\; \delta\sum_i |v_i|\,\|w_{i,\mathcal{B}\times S}\|_1 \;\le\; \delta\sqrt{kL}\sum_i |v_i|\|w_i\| = \delta\sqrt{kL}\,R(\theta).$$

Since $R_{\min} = G/\Delta = G/(\delta\sqrt{kL})$, the ratio follows. The unequal case replaces $kg \le \dots$ by $\sum_j g/\delta_j \le \dots$. □

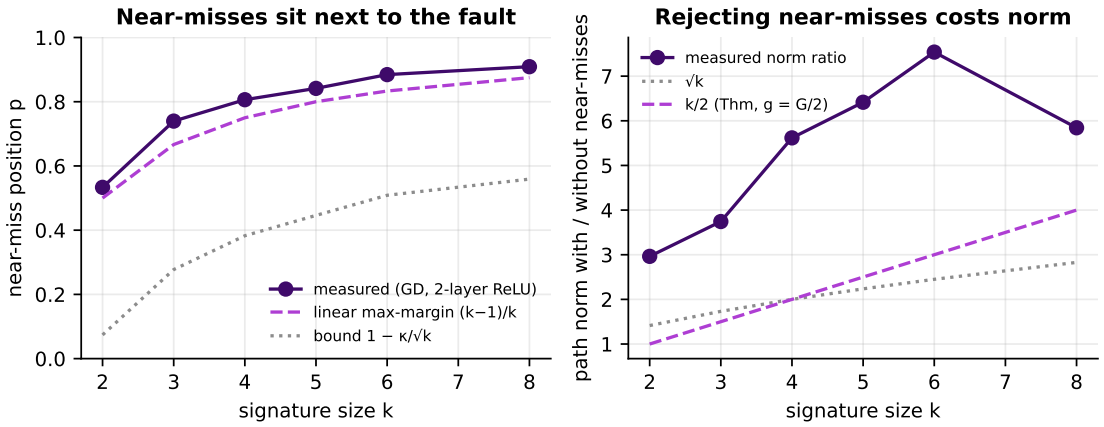


Figure 3: Separated regime, two-layer ReLU networks trained by gradient descent (check C4). Left: where the near-miss lands between normal (0) and fault (1). Right: path-norm increase needed once near-misses are added as negatives.

Rejecting the near-misses down to the normal level ($g = G$) costs at least k times the minimum path norm; pushing them to the midpoint ($g = G/2$) costs at least $k/2$. Check C4 measures ratios of 2.96–7.54 for $k = 2, \ldots, 8$ when near-misses are added (Figure 3).

Proposition 5.5 (Exact linear limit under symmetric data; from Lyu et al. (2021))**.** *If the training set is linearly separable and symmetric, i.e. closed under $(x, y) \mapsto (-x, -y)$ (true in distribution after centring balanced normal/fault data with symmetric noise; exactly after symmetrising the sample), then gradient flow from small initialisation on two-layer leaky-ReLU networks converges in direction to the linear max-margin classifier (Lyu et al., 2021, main theorem). For our prototypes that classifier is $w \propto \mathbf{1}_S$, so near-misses sit at exactly $p = (k-1)/k$. They are classified positive by the max-margin threshold for every $k \geq 3$ and lie on it for $k = 2$.*

Proposition 5.6 (How many near-misses beat weight decay)**.** *Consider noise-free prototypes with masses q at x_0 (label -1), q at x_+ (label $+1$) and p/k at each near-miss (label -1), $2q + p = 1$, in shift-statistic space with equal shifts δ. The objective is $J(\theta) = \sum \text{mass} \cdot \log(1 + e^{-y f_\theta(x)}) + \lambda R(\theta)$, the function-space form of ℓ_2 weight decay by Lemma 5.1. Define*

$$J_{\text{cheap}}(p) = \min_{m_0, m_1 \geq 0} \; q\ell(m_0) + q\ell(m_1) + p\,\ell\Big(-m_1 + \tfrac{m_0+m_1}{k}\Big) + \lambda \tfrac{m_0+m_1}{\delta\sqrt{k}},$$

$$B(p) = \inf_{m, m_0 \in \mathbb{R}} \; q\ell(m) + q\ell(m_0) + \lambda \max\Big(\tfrac{\sqrt{k}\, m_+}{\delta}, \tfrac{|m+m_0|}{\delta\sqrt{k}}\Big),$$

with $\ell(z) = \log(1+e^{-z})$. If $J_{\text{cheap}}(p) < B(p)$, then every minimiser of J flags at least one near-miss ($f(x^{(j)}) > 0$). Hence the near-miss fraction must exceed $p^*(\lambda, k) = \sup\{p : J_{\text{cheap}}(p) < B(p)\}$ before weight-decay training can learn the narrow rule.

Proof. Upper bound. The single unit $f(x) = v\,\text{relu}(\mathbf{1}_S^\top x/\sqrt{k}) - m_0$ with $v = (m_0 + m_1)/(\delta\sqrt{k})$ satisfies $f(x_0) = -m_0$ and $f(x_+) = m_1$. It is linear along the prototypes, so $f(x^{(j)}) = m_1 - (m_0 + m_1)/k$, and its path norm is v. Hence $\min J \leq J_{\text{cheap}}(p)$.

Lower bound. Let f reject every near-miss, $f(x^{(j)}) \leq 0$, and set $m = f(x_+)$, $m_0 = -f(x_0)$. The near-miss losses are nonnegative. Theorem 5.4 with $g = m_+$ gives $R \geq \sqrt{k}\, m_+/\delta$, and Lemma 5.1 gives $R \geq |m + m_0|/(\delta\sqrt{k})$. Hence $J(\theta) \geq B(p)$. If $J_{\text{cheap}}(p) < B(p)$, no function rejecting all near-misses is a minimiser. □

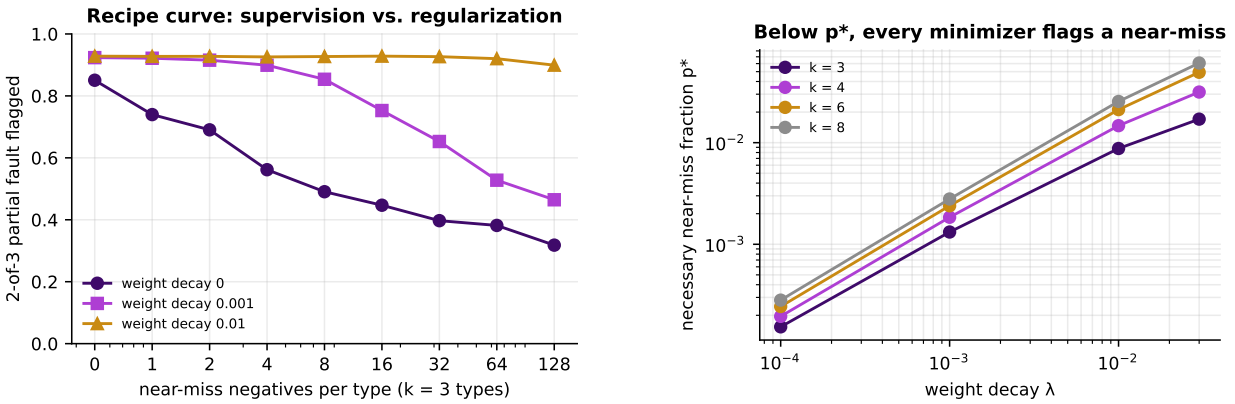


Figure 4: Left: partial-match false alarms vs. near-miss negatives per type, two-layer ReLU network trained by gradient descent with three weight decays (check C5). Right: the necessary near-miss fraction $p^*(\lambda, k)$ of Proposition 5.6 (check C9).

Numerically (check C9, $\delta = 1$): $p^* \approx 0.0002, 0.0013, 0.0088, 0.017$ for $\lambda = 10^{-4}, 10^{-3}, 10^{-2}, 3 \cdot 10^{-2}$ at $k = 3$, and p^* grows with k (Figure 4). The necessary fraction grows roughly linearly in λ for small λ and somewhat more slowly beyond. In finite-noise gradient-descent experiments (C5) the required amount is larger, and at $\lambda = 10^{-2}$ even 128 near-misses per type do not move the partial-match rate. This is a lower bound, not a sufficient condition.

6 Time: what window features and temporal dependence do

Lemma 6.1 (Window statistics are timing-blind)**.** *Let $\psi : \mathbb{R}^T \to \mathbb{R}^q$ be invariant to permutations of the time index (mean, standard deviation, min, max, quantiles), and let $\Psi(X) = (\psi(X_{\cdot 1}), \ldots, \psi(X_{\cdot C}))$. Suppose the noise is exchangeable in time within each channel and independent across channels. If two shift patterns $G, G' \in \mathbb{R}^{T \times C}$ have, channel by channel, the same multiset of entries, then $\Psi(W+G) \stackrel{d}{=} \Psi(W+G')$. Every detector built on Ψ flags simultaneous, staggered and pair-staggered faults with the same per-channel shift durations at identical rates.*

Proof. For each j choose a permutation π_j of $[T]$ with $G'_{\cdot j} = G_{\pi_j(\cdot), j}$. Exchangeability gives $W_{\cdot j} \stackrel{d}{=} W_{\pi_j(\cdot), j}$, and independence across channels makes this hold jointly. Then $(W+G')_{\cdot j} \stackrel{d}{=} (W+G)_{\pi_j(\cdot), j}$ jointly over j, and ψ is permutation-invariant. □

For AR noise or for statistics such as the slope and the last-8 mean, the invariance is approximate. This matches window-statistics models flagging staggered faults at 0.57–1.00 on every target.

Lemma 6.2 (Dispersion features are sign-blind)**.** *If $W \stackrel{d}{=} -W$ and ψ is a dispersion functional ($\psi(-x) = \psi(x)$, e.g. standard deviation or range), then $\psi(W+G) \stackrel{d}{=} \psi(W-G)$. Detectors relying on dispersion flag a fault and its sign-flipped version equally.*

Proof. $\psi(W+G) = \psi(-W-G) \stackrel{d}{=} \psi(W-G)$. □

This is consistent with the fan-coil LSTM flagging wrong-direction changes (0.46–0.83) and the toy gradient-boosting model's reliance on window standard deviation.

Lemma 6.3 (Temporal dependence enters through σ_{eff})**.** *For stationary noise with autocovariance $\gamma(h)$, $\mathrm{Var}(a_j) = c^\top \Gamma c$ with $c = (-\mathbf{1}_B/|B|,\ \mathbf{1}_I/|I|)$ and $\Gamma_{st} = \gamma(s-t)$. For AR(1) with coefficient ϕ and marginal variance σ_x^2, the variance of an L-step mean is $\sigma_x^2 L^{-2} \sum_{s,t} \phi^{|s-t|} \to \sigma_x^2(1+\phi)/((1-\phi)L)$. All results above hold with $\sigma \leftarrow \sigma_{\text{eff}}$. A stickier process lowers D, which lowers both detection and partial false alarms along the same curve (Corollary 3.4). For sample sizes, n consecutive windows count as $n_{\text{eff}} = n(1-\phi)/(1+\phi)$.*

Proof. a_j is the linear functional $c^\top W_{\cdot j}$ plus a constant. The AR(1) limit is the standard series $\sum_{h\in\mathbb{Z}} \phi^{|h|} = (1+\phi)/(1-\phi)$. Check C7 matches the exact formula to within sampling error. □

7 Minimal supervision: what to add and how much

Normal-vs-fault data label only the bottom ($\mathbf{0}$) and top ($\mathbf{1}$) of the pattern lattice. For $k = 3$, 18 monotone rules fit those labels perfectly (20 monotone Boolean functions of three variables minus the two constants), and AND and OR are both among them.

Theorem 7.1 (Channel teaching set)**.** *Let $\mathcal{H} = \{h : \{0,1\}^k \to \{0,1\}$ monotone, $h(\mathbf{0}) = 0,\ h(\mathbf{1}) = 1\}$. A set $\mathcal{L} \subseteq \{0,1\}^k$ labelled by AND identifies AND within $\mathcal{H}$ if and only if $\mathcal{L} \supseteq \{\mathbf{1} - e_j : j \in [k]\}$.*

Proof. ($\Leftarrow$) Any $x \neq \mathbf{1}$ has some $x_j = 0$, so $x \leq \mathbf{1} - e_j$, and monotonicity gives $h(x) \leq h(\mathbf{1} - e_j) = 0$. ($\Rightarrow$) If $\mathbf{1} - e_i \notin \mathcal{L}$, then $h_i(x) = \mathbf{1}\{x \geq \mathbf{1} - e_i\}$ is monotone with $h_i(\mathbf{0}) = 0$ (as $k \geq 2$) and $h_i(\mathbf{1}) = 1$. It equals AND everywhere except at $\mathbf{1} - e_i$, so it is consistent with $\mathcal{L}$ and differs from AND. □

This is the standard boundary characterisation of teaching sets for monotone functions (Goldman and Kearns, 1995); it is included for completeness. The new part is the temporal analogue.

Theorem 7.2 (Temporal teaching set via Helly)**.** *Let each signature channel be shifted on a nonempty interval $I_j \subseteq [0,T)$, and let $\mathcal{G}(I)$ be the overlap graph on S (edge ij iff $I_i \cap I_j \neq \emptyset$).*

(a) (Helly, $d = 1$) $\bigcap_j I_j \neq \emptyset$ if and only if $\mathcal{G}(I) = K_k$.

(b) Consider rules $h(I) = H(\mathcal{G}(I))$ with H monotone under edge inclusion and $H(K_k) = 1$. A labelled set identifies the simultaneity rule $H^*(G) = \mathbf{1}\{G = K_k\}$ within this class if and only if it contains, for every edge e, a negative with overlap graph $K_k - e$.

(c) Each $K_k - e$ is realisable by intervals of equal length (so per-channel shift durations match the target's): for $e = \{a, b\}$ take $I_a = [0,1)$, $I_b = [1,2)$ and $I_j = [\frac{1}{2}, \frac{3}{2})$ otherwise.

(d) A fully staggered negative (empty overlap graph) is the bottom of the lattice. For $k \geq 3$, labelling it 0 does not identify H^*. (For $k = 2$, $K_2 - e$ is the empty graph, so the two notions coincide.)

Proof. (a) Pairwise-intersecting intervals have a common point: with $I_j = [l_j, u_j)$, pairwise intersection gives $\max_j l_j < \min_j u_j$, and any point in between is common. The converse is trivial. (b) Identify G with its edge-indicator vector in $\{0,1\}^{\binom{k}{2}}$. Then H^* is the AND of the edge indicators. Sufficiency: every $G \neq K_k$ is contained in some $K_k - e$, so $H(G) \leq H(K_k - e) = 0$. Necessity: if $K_k - e$ is not labelled, $H_e(G) = \mathbf{1}\{G \supseteq K_k - e\}$ is monotone, satisfies $H_e(K_k) = 1$, and differs from H^* only at $K_k - e$. It is therefore consistent with every other labelled configuration. (c) $I_a \cap I_j = [\frac{1}{2}, 1)$, $I_b \cap I_j = [1, \frac{3}{2})$, $I_j \cap I_{j'} = I_j$, and $I_a \cap I_b = \emptyset$. (d) For $k \geq 3$, $K_k - e$ has at least two edges, so $H(G) = \mathbf{1}\{G \supseteq K_k - e\}$ is monotone, equals 0 on the empty graph, and differs from H^*. □

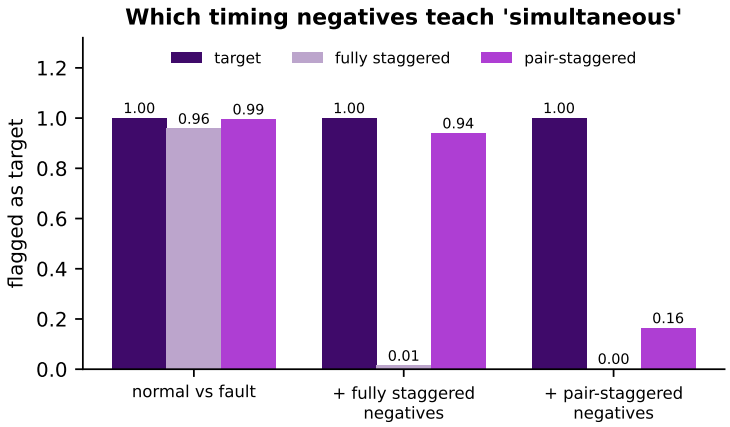


Figure 5: Temporal teaching set (check C6b). A raw-series network trained only on normal vs. fault flags both staggered variants. Fully staggered negatives fix only fully staggered faults. Pair-staggered negatives (the Helly boundary) fix both.

Corollary 7.3 (Total near-miss types)**.** *Consider the two axes separately: the channel axis (subsets $A \subseteq S$ shifted, all simultaneous) with rules monotone in A, and the time axis (all of S shifted, overlap graph G) with rules monotone in G. A rule is any pair of such rules agreeing at the shared target. Identification then requires and is achieved by the k channel near-misses and the $\binom{k}{2}$ one-edge-missing patterns, $k(k+1)/2$ types in total.*

Proof. The two axes share only the target configuration, where every rule equals 1, so the constraints decouple. Apply Theorem 7.1 on the channel axis and Theorem 7.2(b) on the time axis. □

Remark 7.4 (Correction to an earlier version). An earlier version claimed the same count for rules monotone in the *joint* order $(A, G) \preceq (A', G')$. That necessity claim is false. The channel near-miss $(S \setminus \{j\}, K_{S\setminus\{j\}})$ lies below $(S, K_k - \{i, j\})$, so under joint monotonicity the $\binom{k}{2}$ pair-staggered negatives alone suffice. The per-axis statement above is the correct one for learners that need not respect the joint order.

Check C6b confirms the prediction of Theorem 7.2(d) with a raw-series network trained by gradient descent. Fully staggered negatives fix fully staggered faults ($0.96 \to 0.01$) but leave pair-staggered ones at 0.94. Pair-staggered negatives fix both (0.00 and 0.16). In our real-data pipeline the "staggered negatives" were fully staggered, which explains why they did not generalise; the pipeline now has pair-staggered negatives.

Proposition 7.5 (How many examples per near-miss type)**.** *For a learner with the intersection–union structure built in (per-channel thresholds, then AND), let channel j have a level ν_j in its near-miss and $\nu_j + \Delta_j$ in the target, with noise σ_j (effective, Lemma 6.3). Set $\hat{c}_j$ to the midpoint of the sample means from n near-misses of type j and n targets. If*

$$n \;\geq\; \frac{32 \max_j \sigma_j^2}{\min_j \Delta_j^2} \log \frac{4k}{\eta},$$

then with probability at least $1 - \eta$, every threshold is within $\Delta_j/4$ of its midpoint. The learned rule then flags near-miss j with probability at most $\Phi(-\Delta_j/(4\sigma_j))$ and misses the target with probability at most $\sum_j \Phi(-\Delta_j/(4\sigma_j))$. For windows drawn consecutively from an AR(1) process, replace n by n_{eff}.

Proof. A mean of n independent $N(\mu, \sigma^2)$ samples deviates by more than t with probability at most $2e^{-nt^2/(2\sigma^2)}$. The midpoint error is at most the larger of the two mean errors, so with $t = \Delta_j/4$ the failure probability for channel j is at most $4e^{-n\Delta_j^2/(32\sigma_j^2)}$; a union bound over j gives the condition. On success $\hat{c}_j \geq \nu_j + \Delta_j/4$, so near-miss j exceeds it with probability at most $\Phi(-\Delta_j/(4\sigma_j))$; the target bound is symmetric plus a union bound. Under dependence the variance of the mean is inflated by $(1+\phi)/(1-\phi)$ (Lemma 6.3). □

Using near-misses rather than normal data matters when sensors are coupled. In a real near-miss, channel j's level ν_j is shaped by the other faulty channels and need not equal its normal level.

8 Answer: why, what is missing, and the recipe

Why. In both regimes the detector learned from normal-vs-fault data is additive or Lipschitz-small across channels and time. With overlapping classes this is the Bayes-optimal training rule (Theorem 3.1) and gradient descent converges to it. With separated classes, implicit bias selects small-norm solutions, which cannot separate the target from near-misses without paying k times the norm (Theorems 5.2–5.4). The narrow rule is never favoured, and the stronger the target, the worse the partial-match false alarms (Corollary 3.4).

What is missing. Not data volume or class balance (Theorem 3.1(ii)–(iii)), but one of the following: (1) the right null: calibrate against normal *and* near-misses (Theorem 4.1); (2) the right supervision: k channel and $\binom{k}{2}$ temporal near-misses (Theorems 7.1–7.2); (3) the right inductive bias: capped per-channel evidence or an intersection–union structure (Proposition 4.2).

Recipe.

1. Estimate the signature S from fault vs. normal data (top-k standardised shifts).
2. Either build the conjunction in (per-channel capped evidence, then AND) or train a flexible model with near-miss negatives of all $k + \binom{k}{2}$ types, using *pair*-staggered rather than fully staggered timing negatives.
3. Amounts: structured learner, $n \geq 32(\sigma_{\text{eff}}/\Delta)^2 \log(4k/\eta)$ per type (Proposition 7.5). Black-box with weight decay λ, near-miss fraction above $p^*(\lambda, k)$, which grows with λ (Proposition 5.6); in practice more (Figure 4).
4. Calibrate the alarm threshold so that near-misses, not only normal windows, stay below the budget.

9 Numerical checks

All checks are in `theory/theory_checks.py` (NumPy and SciPy; gradient-descent learners where the claim is about gradient descent). Output is in `theory/checks_out/`.

C1 GD logistic regression, $k = 3$, $\delta/\sigma = 1.5$: predicted vs. measured range over $n \in \{2000, 20000\}$ and prevalence $\in \{0.1, 0.5, 0.9\}$: CAP 0.607 (0.585–0.610), P2 0.276 (0.250–0.282), P1 0.072 (0.064–0.072), BIG1 0.607 (0.544–0.619), UP 0.000 (0.000), OFF 0.010 (0.008–0.012). A two-layer network follows the trade-off curve from its own detection rate (P2 0.208 predicted vs. 0.203).

C2 Paradox: as δ/σ goes $0.5 \to 4$, measured P2 goes $0.039 \to 0.985$ (predicted $0.040 \to 0.989$).

C3 Composite null: additive detectors calibrated on normal have size 0.49–1.00 on near-misses; under unbounded overlap every additive detector has size 1.00. The intersection–union detector has size 0.010 and power 0.051/0.423/0.866/0.989 at $\delta/\sigma = 2/3/4/5$, matching $\Phi(\delta/\sigma - z)^3$; the bounded-null additive detector has 0.121/0.276/0.493/0.712; the capped GAM is valid with power 0.00/0.37/0.72/0.91 (Figure 2).

C4 Gradient-descent ReLU nets, separated classes: near-miss position 0.53–0.91 ($\geq$ linear $(k-1)/k$ and $\geq$ the bound), $\kappa \approx 1.2$–1.3, near-misses flagged 98–100%. Adding near-misses raises the normalised path norm by 2.96–7.54× ($\geq \sqrt{k}$; Theorem 5.4 gives $\geq k\, g/G$).

C5 P2 false alarms vs. near-misses per type ($0 \to 128$): $0.85 \to 0.32$ without weight decay; $0.92 \to 0.46$ at $\lambda = 10^{-3}$; flat at $0.93 \to 0.90$ at $\lambda = 10^{-2}$.

C6 (a) Window statistics (mean, sd, min, max) with i.i.d. noise: target, fully staggered and pair-staggered (equal durations) flagged at 0.276/0.274/0.270, as Lemma 6.1 requires. (b) Helly: see Section 7 and Figure 5.

C7 AR(1) variance formula vs. simulation: agreement within sampling error.

C8 Toy slide 12 predicted from detection rate alone: MLP/LSTM P2 0.72 vs. 0.73, P1 0.19 vs. 0.19/0.20.

C9 Necessary near-miss fraction $p^*(\lambda)$, as in Proposition 5.6.

Pipeline experiments on the toy replica (slide-7 data, LSTM and five other models)

Same code path as the real-data runs (`run_toy.py`, `reference_results/toy_theory`). Noisy, overlapping-class regime.

Prediction	Toy result (LSTM unless stated)	Verdict
Partial-match law (Cor. 3.4), from CAP + single-channel rates	P2 predicted 0.58, measured 0.58	holds
Paradox: stronger fault ⇒ more partial alarms	scale ×0.25/0.5/0.75/1/1.5: CAP 0.09/0.27/0.74/0.96/1.00, P2 0.05/0.10/0.34/0.59/0.94; law 0.05/0.12/0.37/0.66/0.89. Same pattern for all five models.	holds
Capped evidence ignores one loud channel (Prop. 4.2)	BIG1 flagged, no fix: capped CI 0.08 vs uncapped CI 0.99	holds
Capped CI is fixable, uncapped is not	after all-but-one negatives P2: capped 0.37 → 0.22, uncapped 0.72 → 0.35	partial
Recipe: more near-misses ⇒ fewer partial alarms	P2 by near-misses per type 0/30/100/300/1000: 0.58/0.47/0.35/0.18/0.16 (CAP 0.95 → 0.86); with weight decay 10^{-3}: 0.53 → 0.14	holds; needs ∼300/type here
Helly: pair-staggered negatives teach timing, fully staggered do not	PAIRSTAG flagged: 0.30 before; 0.17 with fully staggered negatives; 0.06 with pair-staggered	direction holds, effect smaller than in C6b

10 How this revises the earlier proposal (Lemmas 1–4)

The earlier ICLR draft framed the problem as gradient descent choosing between a narrow k-junta and a broad correlate, with four lemmas: effective sample size under mixing (L1), max-margin preference for the broad rule (L2), flatness and stability (L3), and a predicted gap scaling as $\sqrt{(1+\rho)/(1-\rho)}/\sqrt{n}$ (L4). Its synthetic results confirmed L1, found the learned rule to be *additive rather than either/or*, could not establish the L4 scaling (no correlation over five seeds), found a weight-norm confound in L3, and found that a KL-leash fixed the probed trigger but did not generalise to an unprobed one. The present theory explains each of these.

Earlier	What the theory now says	Why the earlier result came out as it did
L1: $n_{\text{eff}} = n(1-\rho)/(1+\rho)$	Kept (Lemma 6.3). Dependence enters only through σ_{eff} and n_{eff} (Proposition 7.5).	Confirmed. Estimating n_{eff} from a continuous channel, not the sparse label, remains correct.
L2: GD prefers the broad rule by max margin	Replaced. In the overlap regime there is no margin: the broad (additive) rule is the Neyman–Pearson optimum of the training task and consistent learners converge to it (Theorem 3.1). In the separated regime, any norm-bounded network places near-misses next to the fault (Theorem 5.2).	The hard-margin precondition failed on noisy data. The new statements do not need it.
L3: narrow rule is sharper / less stable	Replaced by a norm statement: rejecting the k near-misses needs $\geq k\,g/G$ times the minimum path norm (Theorem 5.4).	The "confound" (the narrow model had larger weight norm) is the predicted effect itself. Normalising noise by the norm would remove what the theorem predicts.
L4: gap $\propto \sqrt{(1+\rho)/(1-\rho)}/\sqrt{n}$	No such law. The shortcut is the $n \to \infty$ limit (Theorem 3.1(iii)), so the gap does not close with n_{eff}. Slower mixing lowers D, which moves the detector *along* the power–specificity curve: lower detection *and* fewer partial false alarms (Corollary 3.4).	The predicted scaling was not observed (no correlation over five seeds). This is what the theory expects.
"Additive, not exclusive" (earlier results, Sec. 6)	Exactly Theorem 3.1(i) and Theorem 3.3: the learned score is a sum of per-channel evidence that includes the true channels.	—
KL-leash fixes the probed trigger, not a novel one	Identification needs *every* maximal near-miss type (Theorems 7.1, 7.2). Penalising one direction teaches one type.	Expected. The prediction is that a probe set covering all k channel and $\binom{k}{2}$ pair-staggered types generalises to every partial fault on the signature.

The research question therefore changes from "how much does dependence widen the gap?" to "what supervision or inductive bias makes the conjunction identifiable, and how much of it is needed?". The answer to the second question is in Section 7 and Proposition 5.6.

11 Relation to prior work and what is new

Known, and used here. Neyman–Pearson; intersection–union tests (Berger, 1982); teaching sets of monotone functions (Goldman and Kearns, 1995) and contrastive or minimal-pair examples (Mansouri et al., 2025); implicit bias of gradient descent (Soudry et al., 2018; Lyu and Li, 2020; Ji and Telgarsky, 2020; Lyu et al., 2021); simplicity bias (Shah et al., 2020; Morwani et al., 2023); gradient starvation (Pezeshki et al., 2021); feature availability vs. predictivity (Hermann et al., 2024); shortcut learning (Geirhos et al., 2020); underspecification (D'Amour et al., 2022); open-set fault diagnosis.

Plausibly new (to be confirmed by a full literature review).

1. Viewing the multi-sensor shortcut as the *Neyman–Pearson-optimal detector of the training task*, together with the closed-form partial-overlap law. Prior shortcut theory studies one core and one spurious feature; here there is no spurious feature, and the failure is the logical structure (sum vs. AND) of the correct features.

2. The *detectability paradox* and the power–specificity curve, which predict partial-match false alarms from a model's detection rate alone.

3. Fleet false alarms as a *composite-null* problem: the $\sqrt{k}$ penalty and the zero-power result for additive and channel-independent detectors, and capped evidence as necessary and sufficient. This links to the channel-independence design of current time-series foundation models.

4. The near-miss position bound and the factor-k path-norm cost, giving a mechanism for *why* norm-seeking training does not find the conjunction, and a necessary near-miss fraction that grows with weight decay.

5. The *temporal* teaching set via Helly's theorem ($\binom{k}{2}$ pair-staggered types) and the prediction, confirmed numerically, that fully staggered negatives do not teach simultaneity.

What reviewers will ask. (i) Global optimality: gradient descent reaches KKT points, not necessarily the global max-margin solution. Theorems 5.2–5.4 hold for *any* network in terms of its measured κ, which sidesteps this, and C4 measures κ. (ii) Gaussianity: Theorem 3.1 extends to exponential families; Section 5 needs no distributional assumption. (iii) Simulated benchmarks: add multi-run Tennessee Eastman and fleet data. (iv) The signature is estimated: report robustness over k and selection thresholds.

12 Open problems

1. A convergence theorem for gradient descent on two-layer ReLU nets with *non-symmetric* normal/fault data showing $p(x^{(j)}) \geq (k-1)/k$ (C4 measures p above this line).

2. The exact partial-overlap law for the unknown-onset scan statistic.

3. Explaining the fan-coil LSTM's direction-free ("something changed") rule beyond Lemma 6.2.

4. Signature recovery with dependent windows and its effect on the recipe.

5. Tightness of $p^*(\lambda)$: closing the gap between the necessary fraction and what gradient descent needs in practice (C5 vs. C9).

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